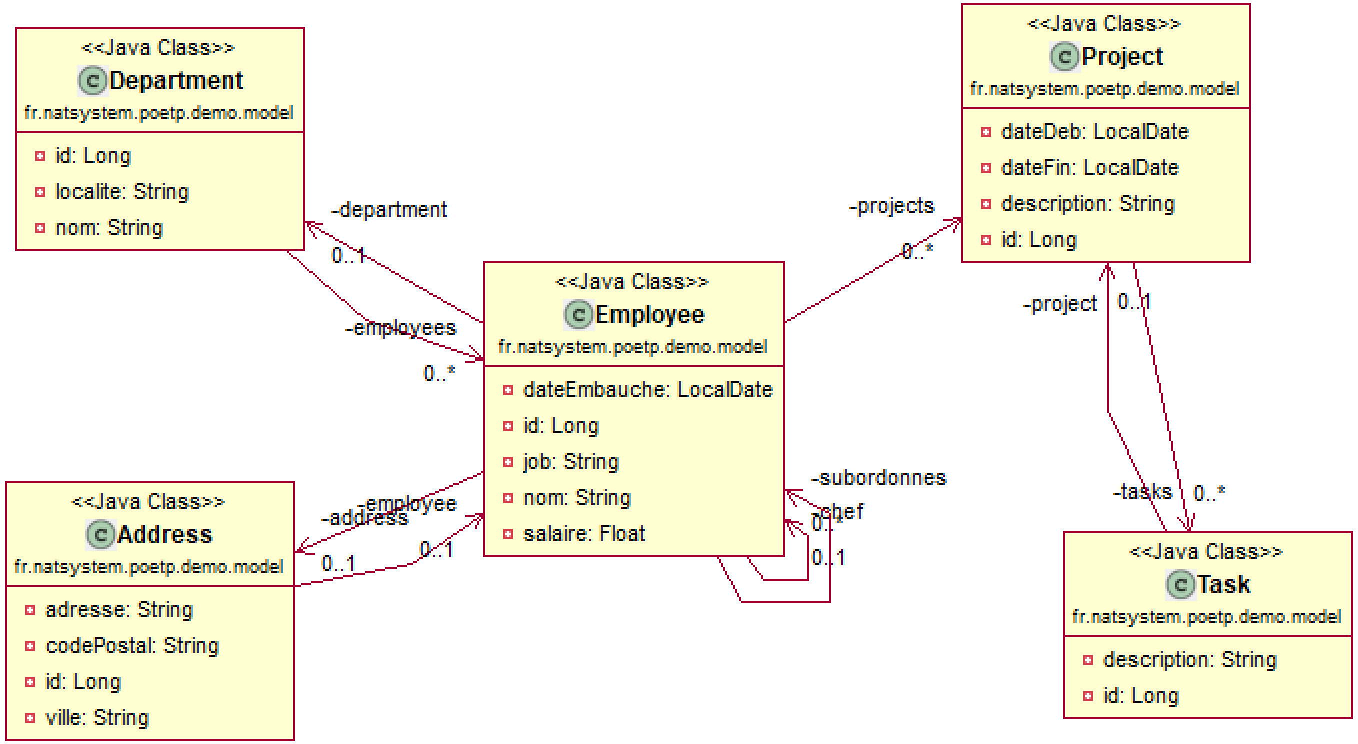
Module Frameworks : test de fin

Le modèle de données

Diagramme de classes du test



Les tables de BDD
et les classes d'Entity

1. Des Employés

- Table : Z_EMPLOYEE
    - Primary key "EMP_ID"
    - Initialisée par la séquence "Z_EMP_SQ"

	COLUMN_NAME	DATA_TYPE	NULLABLE
1	EMP_ID	NUMBER	No
2	EMP_NOM	VARCHAR2(30 BYTE)	Yes
3	EMP_JOB	VARCHAR2(30 BYTE)	Yes
4	EMP_BOSS_RF	NUMBER	Yes
5	EMP_DATE_EMBAUCHE	DATE	Yes
6	EMP_SALAIRE	NUMBER(6,2)	Yes
7	EMP_DEPT_REF	NUMBER	Yes

- Classe java : Employee

La table Z_EMPLOYEE

```sql
CREATE SEQUENCE Z_EMP_SQ;

CREATE TABLE Z_EMPLOYEE(
    EMP_ID NUMBER NOT NULL PRIMARY KEY,
    EMP_NOM VARCHAR2(30),
    EMP_JOB VARCHAR2(30),
    EMP_BOSS_RF NUMBER,
    EMP_DATE_EMBAUCHE DATE,
    EMP_SALAIRE NUMBER(6, 2),
    EMP_DEPT_REF NUMBER
);
```

La classe Employee

```
public class Employee {
    private Long id;
    private String nom;
    private String job;
    private LocalDate dateEmbauche;
    private Double salaire;

    // entités liées
    private Address address;
    private Employee chef;
    private Set<Employee> subordonnes;
    private Department department;
    private Set<Project> projects;

}
```

Données

	EMP_ID	EMP_NOM	EMP_JOB	EMP_BOSS_RF	EMP_DATE_EMBAUCHE	EMP_SALAIRE	EMP_DEPT_REF
1	1	KING	PRESIDENT	(null)	17/11/1981 00	5000	1
2	2	JONES	MANAGER	1	02/04/1981 00	2975	2
3	3	SCOTT	ANALYST	2	09/12/1982 00	3000	2
4	4	ADAMS	CLERK	5	12/01/1983 00	1100	2
5	5	FORD	ANALYST	2	03/12/1981 00	3000	2
6	6	SMITH	CLERK	5	17/12/1980 00	800	2
7	7	BLAKE	MANAGER	1	01/05/1981 00	2850	3
8	8	ALLEN	SALESMAN	7	20/02/1981 00	1600	3
9	9	WARD	SALESMAN	13	22/02/1981 00	1250	3
10	10	MARTIN	SALESMAN	13	28/09/1981 00	1250	3
11	11	TURNER	SALESMAN	7	08/09/1981 00	1500	3
12	12	JAMES	CLERK	11	03/12/1981 00	(null)	3
13	13	CLARK	MANAGER	1	09/06/1981 00	2450	1
14	14	MILLER	CLERK	13	23/01/1982 00	1300	1

2. Des Départements
(un employé est rattaché à un départment)

- Table : Z_DEPARTMENT
  - Primary key "DEP_ID"
  - Initialisée par la séquence "Z_DEP_SQ"

	COLUMN_NAME	DATA_TYPE	NULLABLE
1	DEP_ID	NUMBER	No
2	DEP_NOM	VARCHAR2(30 BYTE)	Yes
3	DEP_LOC	VARCHAR2(30 BYTE)	Yes

- Classe java : Department

La table Z_DEPARTMENT

```sql
CREATE SEQUENCE Z_DEP_SQ;

CREATE TABLE Z_DEPARTMENT(
    DEP_ID NUMBER NOT NULL PRIMARY KEY,
    DEP_NOM VARCHAR2(30),
    DEP_LOC VARCHAR2(30)
);
```

La classe Department

```java
public class Department {
   private Long id;
    private String nom;
    private String localite;

    // entités liées
    private Set<Employee> employees;
}
```

Données

	DEP_ID	DEP_NOM	DEP_LOC
1	1	ACCOUNTING	NEW YORK
2	2	RESEARCH	DALLAS
3	3	SALES	CHICAGO
4	4	OPERATIONS	BOSTON

3. Des Adresses postales

(un employé a une adresse postale)

- ❖Table : Z_ADDRESS
  - ❖Primary key "ADR_ID"
  - ❖Initialisée par la séquence "Z_ADR_SQ"

1	ADR_ID	NUMBER	No
2	ADR_ADRESSE	VARCHAR2(100 BYTE)	Yes
3	ADR_VILLE	VARCHAR2(30 BYTE)	Yes
4	ADR_CODE_POSTAL	VARCHAR2(5 BYTE)	Yes
5	ADR_EMP_REF	NUMBER	Yes

- ❖Classe java : Address

La table Z_ADDRESS

```sql
CREATE SEQUENCE Z_ADR_SQ;

CREATE TABLE Z_ADDRESS(
    ADR_ID NUMBER NOT NULL PRIMARY KEY,
    ADR_ADRESSE VARCHAR2(100),
    ADR_VILLE VARCHAR2(30),
    ADR_CODE_POSTAL VARCHAR2(5),
    ADR_EMP_REF NUMBER
);
```

La classe Address

```java
public class Address {

    private Long id;
    private String adresse;
    private String ville;
    private String codePostal;

    // entités liées
    private Employee employee;
}
```

Données

	ADR_ID	ADR_ADRESSE	ADR_VILLE	ADR_CODE_POSTAL	ADR_EMP_REF
1	1	1 RUE DE LA GARE	PARIS	75000	1

4. Des Projets
(un employé participe à des projets)

- ❖Table : Z_PROJECT
  - ❖Primary key "PRJ_ID"
  - ❖Initialisée par la séquence "Z_PRJ_SQ"

	COLUMN_NAME	DATA_TYPE	NULLABLE
1	PRJ_ID	NUMBER	No
2	PRJ_DESCRIPTION	VARCHAR2(100 BYTE)	Yes
3	PRJ_DATE_DEB	DATE	Yes
4	PRJ_DATE_FIN	DATE	Yes

- ❖Classe java : Project

La table Z_PROJECT

```sql
CREATE SEQUENCE Z_PRJ_SQ;

CREATE TABLE Z_PROJECT(
    PRJ_ID NUMBER NOT NULL PRIMARY KEY,
    PRJ_DESCRIPTION VARCHAR2(100),
    PRJ_DATE_DEB DATE,
    PRJ_DATE_FIN DATE
);
```

La classe Project

```java
public class Project {
    private Long id;
    private String description;
    private LocalDate dateDeb;
    private LocalDate dateFin;

    // entités liées
    private Set<Task> tasks;
}
```

Données

	PRJ_ID	PRJ_DESCRIPTION	PRJ_DATE_DEB	PRJ_DATE_FIN
1	1	Development of Novel Magnetic Suspension System	01/01/2006 00	13/08/2007 00
2	2	Research on thermofluid dynamics in Microdroplets	22/08/2006 00	20/03/2007 00
3	3	Foundation of Quantum Technology	24/02/2007 00	31/07/2008 00
4	4	High capacity optical network	01/01/2008 00	(null)

5. Des Tâches
(un projet est décomposé en tâches)

- ❖Table : Z_TASK
  - ❖Primary key "TSK_ID"
  - ❖Initialisée par la séquence "Z_TSK_SQ"

	COLUMN_NAME	DATA_TYPE	NULLABLE
1	TSK_ID	NUMBER	No
2	TSK_PRJ_REF	NUMBER	Yes
3	TSK_DESCRIPTION	VARCHAR2(100 BYTE)	Yes

- ❖Classe java : Task

La table Z_TASK

```
CREATE SEQUENCE Z_TSK_SQ;

CREATE TABLE Z_TASK(
    TSK_ID NUMBER NOT NULL PRIMARY KEY,
    TSK_PRJ_REF NUMBER,
    TSK_DESCRIPTION VARCHAR2(100)
);
```

La classe Task

```java
public class Task {
    private Long id;
    private String description;

    // entités liées
    private Project project;
}
```

Données

	TSK_ID	TSK_PRJ_REF	TSK_DESCRIPTION
1	1	1	Prj1: Develop feature A
2	2	1	Prj1: Develop feature B
3	3	2	Prj2: Develop feature A
4	4	3	Prj3: Develop feature A
5	5	3	Prj3: Develop feature B
6	6	4	Prj4: Develop feature A

6. Participation des Employés aux Projets

(un employé participe à 0..N projets)
(un projets occupe 0..N employés)

- Table : Z_PROJECT_PARTICIPATION
  - Primary key "PAR_ID"
  - Initialisée par la séquence "Z_PAR_SQ"

	COLUMN_NAME	DATA_TYPE	NULLABLE
1	PAR_ID	NUMBER	No
2	PAR_PRJ_REF	NUMBER	Yes
3	PAR_EMP_REF	NUMBER	Yes
4	PAR_DATE_DEB	DATE	Yes
5	PAR_DATE_FIN	DATE	Yes
6	PAR_ROLE	VARCHAR2(30 BYTE)	Yes

- Classe java : aucune - doit intervenir dans un lien @ManyToMany

La table Z_PROJECT_PARTICIPATION

```sql
CREATE SEQUENCE Z_PAR_SQ;

CREATE TABLE Z_PROJECT_PARTICIPATION(
    PAR_ID NUMBER NOT NULL PRIMARY KEY,
    PAR_PRJ_REF NUMBER,
    PAR_EMP_REF NUMBER,
    PAR_DATE_DEB DATE,
    PAR_DATE_FIN DATE,
    PAR_ROLE VARCHAR2(30)
);
```

Données

	PAR_ID	PAR_PRJ_REF	PAR_EMP_REF	PAR_DATE_DEB	PAR_DATE_FIN	PAR_ROLE
1	1	1	4	01/01/2020 00	(null)	CLERK
2	2	1	6	15/03/2022 00	(null)	CLERK
3	3	2	12	02/02/2023 00	(null)	CLERK

Contraintes référentielles entre tables

```sql
ALTER TABLE Z_EMPLOYEE ADD (
    CONSTRAINT FK_EMP_BOSS FOREIGN KEY (EMP_BOSS_RF) REFERENCES Z_EMPLOYEE (EMP_ID),
    CONSTRAINT FK_EMP_DEPT FOREIGN KEY (EMP_DEPT_REF) REFERENCES Z_DEPARTMENT (DEP_ID)
);
```

```sql
ALTER TABLE Z_ADDRESS ADD (
    CONSTRAINT FK_ADR_EMP FOREIGN KEY (ADR_EMP_REF) REFERENCES Z_EMPLOYEE (EMP_ID)
);
```

```sql
ALTER TABLE Z_PROJECT_PARTICIPATION ADD (
    CONSTRAINT FK_PAR_EMP FOREIGN KEY (PAR_EMP_REF) REFERENCES Z_EMPLOYEE (EMP_ID),
    CONSTRAINT FK_PAR_PRJ FOREIGN KEY (PAR_PRJ_REF) REFERENCES Z_PROJECT (PRJ_ID)
);
```

```sql
ALTER TABLE Z_TASK ADD (
    CONSTRAINT FK_TSK_PRJ FOREIGN KEY (TSK_PRJ_REF) REFERENCES Z_PROJECT (PRJ_ID)
);
```